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Objective: Implement the file loader and get the pipeline working.

Attempts Made:

With the queue and graph done I moved on to loader.h. It needed to do two things: load initial.txt into the adjacency list and parse updates.txt into the adds and delays vectors. The format was pretty readable so the parsing logic itself wasn't hard to write, but I ran into two bugs before getting a correct run.

First one was a segfault right on startup. I was sizing the graph from the vertex count in the file header but hadn't added comment skipping yet, so the parser just read the # line as the first line and allocated a graph of size 1. Any edge with src above 0 wrote out of bounds and crashed.

Fixed that and the program ran to completion but the weight sum came out wrong. Added some print statements to trace where the bad values were coming from and found that the # header lines in both files were going through the numeric parser. src, dst, and weight were all getting wrong values which threw off the total.

Solutions/Workarounds:

I wrote find_max_vertex() to pre-scan both files and get the right graph size before allocating anything. Added line[0] == '#' at the top of every parsing loop. After both fixes the program output weight sum 834418737 and CORRECT.

Learning Outcome:

Pre-scanning before allocating is the same pattern I used for CSR construction back in Phase 2a. The comment line bug was a good reminder that dataset files usually have headers that need to be filtered before you touch any of the numbers.